

SECTION A: STRUCTURED QUESTIONS

Answer **ALL** the questions in this section.

QUESTION 1

Transferrin is a major plasma glycoprotein involved in iron transport. It has a very high affinity for iron, with each transferrin molecule binding two Fe^{3+} ions to form a ferrotransferrin iron-protein complex.

- (a) Based on the above information and your knowledge of the functional role of transferrin, describe the importance of transferrin to mammalian physiology. [2]
- (b) The iron-binding sites in transferrin, found between two globular lobes of the protein, consist of amino acid residues capable of interacting with Fe^{3+} . Identify, with explanation, any two of the following amino acid residues that could possibly be located at the iron-binding sites: Asp, Cys, His, Leu, Phe and Tyr. [2]
- (c) Chicken egg white also contains a type of transferrin called ovotransferrin. Ovotransferrin, like human transferrin, binds iron which functions as an important cofactor. This give rises to one main form of immune defense against microorganisms such as bacteria.

Explain how ovotransferrin serves its role as a form of immune protection, and suggest why the eggs turn bad when washed in iron-containing water from rusty iron pipes even with the presence of ovotransferrin. [2]

- (d) Transferrin has been used as a fusion partner to stabilise and improve the half-life of certain therapeutically important small biomolecules such as glucagon-like peptide 1, which is typically injected into the bloodstream to increase insulin secretion in diabetic patients.
- (i) Explain why an intervening linker peptide is usually engineered to be part of the fusion construct. [1]
- (ii) Describe a relevant experiment to allow for a high level of purification of these fusion constructs. [2]
- (iii) Suggest why the ability of the transferrin fusion construct in improving the half-life of small biomolecules such as glucagon-like peptide 1 is advantageous to the patient. [1]

[Total: 10 marks]

QUESTION 2

Four different monoclonal antibodies TK1, TK15, TK37 and TK47 were raised against the *Xenopus* protein Orc1, which is a component of the DNA replication origin recognition complex (ORC).

- (a) Relate the structure of an antibody molecule to its antigenic specificity. [3]

The four antibodies are used to immunopurify other members of the ORC. To decide which of these antibodies is best suited for this purpose, they are covalently attached to beads, which are then incubated with a *Xenopus* egg extract. The beads are next spun down and washed carefully before the bound proteins are solubilised with SDS. The solubilised proteins are separated via SDS-PAGE and stained as shown in **Fig. 2**.

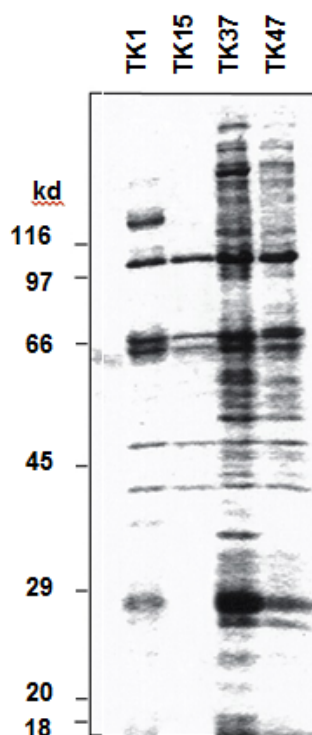


Fig. 2

- (b) (i) With the use of arrows, indicate on **Fig. 2**, all the bands which arise from the proteins in the ORC. [1]
- (ii) Account for your answer in **b(i)**. [1]
- (iii) State the best antibody to study Orc1. [1]

- (c) With reference to **Fig. 2**, explain the different band profiles obtained with each antibody. [2]
- (d) Describe an experiment that allows for the visualisation of the Orc1 band on the gel. [2]

[Total: 10 marks]

QUESTION 3

Typical coiled-coil structures in proteins such as α -keratin and collagen consist of associations between amphipathic α -helices.

- (a) State what is meant by the term *amphipathic* in the context of an α -helix. [1]

Amphipathic α -helices are also known to occur in vertebrate muscle myoglobin, which is made up of eight α -helices **A** to **H**.

An investigator wishes to determine if helix **E**, with the following 19-amino acid long sequence:

Ser-Glu-Asp-Leu-Lys-Lys-His-Gly-Ala-Thr-Val-Leu-Thr-Ala-Lys-Gly-Gly-Ile-Leu

is an amphipathic helix, by arranging the amino acids in order of their position in the helix **E** sequence (shown above) around a helix-wheel plot shown in **Fig. 3.1**.

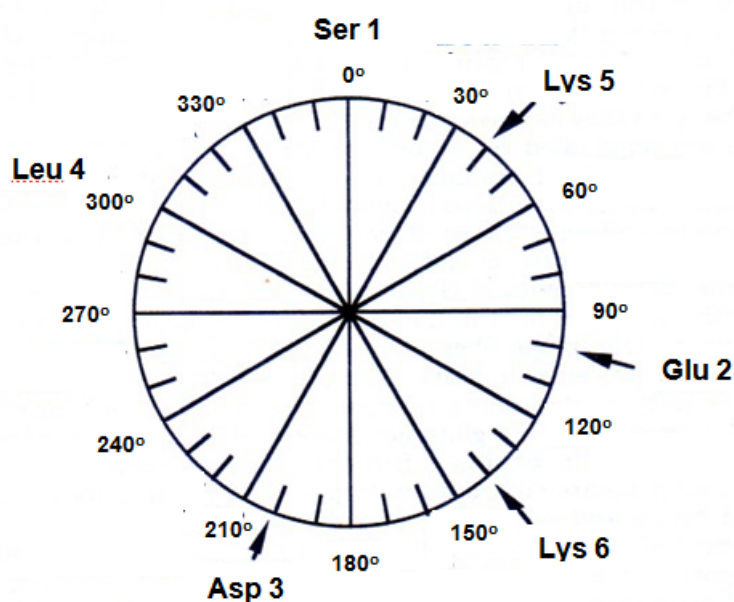


Fig. 3.1

In **Fig. 3.1**, the circle represents the α -helix as viewed from one end. The first five residues of helix **E** – Ser, Glu, Asp, Leu and Lys – are plotted.

- (b) The angle between successive amino acid residues plotted in **Fig. 3.1** is 100° . Briefly explain why. [1]
- (c) Complete the helical wheel plot in **Fig. 3.1** for helix **E**. Use your helical wheel plot to justify if helix **E** is an amphipathic helix, circling the relevant residues to support your answer. [3]
- (d) Based on your knowledge of myoglobin, suggest the structural significance in myoglobin of your conclusion about helix **E** in (c). [1]

The molecular structure of myoglobin can be determined by nuclear magnetic resonance (NMR) spectroscopy. **Fig. 3.2** shows the NMR spectra of myoglobin when bound to oxygen (MbO_2) and carbon monoxide (MbCO).

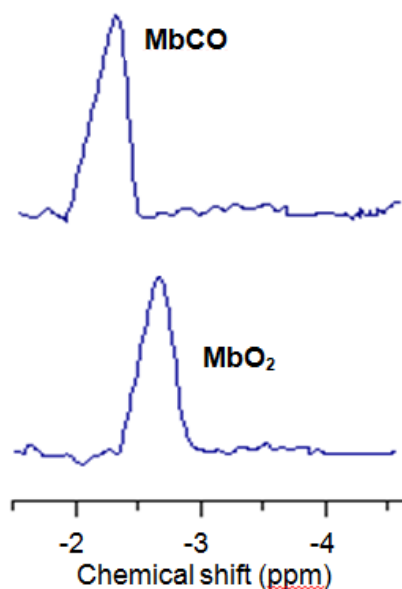


Fig. 3.2

- (e) Outline the purpose of detecting chemical shifts in NMR. [1]
- (f) Explain the effect of ligand binding on the NMR spectra. [1]
- (g) Suggest what the decrease in the peak height of the MbO_2 spectrum represents and therefore the associated medical significance. [2]

[Total: 10 marks]

QUESTION 4

Signal transduction processes mediated by G protein signalling pathways are centred around heterotrimeric G proteins and G protein-coupled receptors (GPCRs).

Fig. 4.1 shows the cross-section of the transmembrane α -helices of a GPCR from the intracellular (IL) perspective. GPCR activation occurs when its ligand binds to the extracellular ligand (EL) - binding domain. Helices 1-7 are transmembrane helices; helix 8 is wholly cytoplasmic and serves as the binding site for the $G_{\beta\gamma}$ subunits. The G_{α} subunit is represented by a star and the shaded portions of helices 3, 5, 6 and 7 represent the G_{α} binding site.

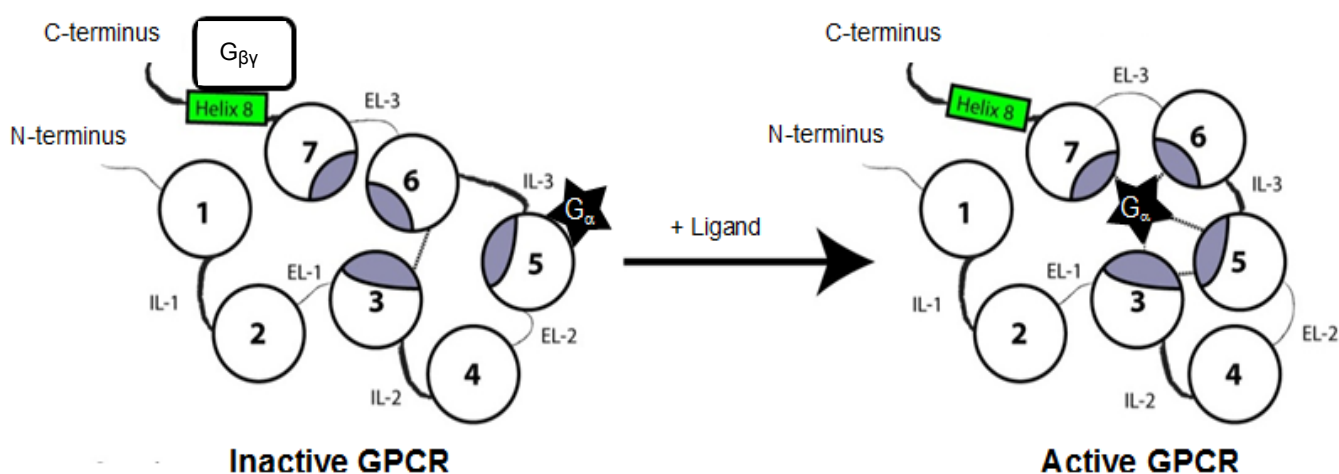


Fig. 4.1

- (a) With reference to **Fig. 4.1**, describe how the conformational change of the GPCR upon ligand binding allows for the coupled G protein to exert its downstream signalling effects. [2]

One such signalling pathway involving GPCRs utilises the β -adrenergic receptor, of which there are three subtypes (β_1 , β_2 and β_3) found in different tissue types. These subtypes can produce different effects in their tissues of origin. All subtypes of the β -adrenergic receptor have adrenaline as their physiological ligand, and act via G_s -type G proteins to stimulate adenylyl cyclase.

- (b) Give one reason why the different subtypes of the β -adrenergic receptors are able to produce different effects in their tissues of origin upon binding of their physiological ligand.

[1]

The β_1 -adrenergic receptor is found in heart muscle tissue, and functions to increase the heart rate upon binding of adrenaline. Carvedilol and metoprolol are two different drugs that act on the β_1 -adrenergic receptor. The effects of the two drugs are shown in **Fig. 4.2**. Baseline activity, the activity of the heart muscle in the presence of adrenaline, is arbitrarily given a value of 100%.

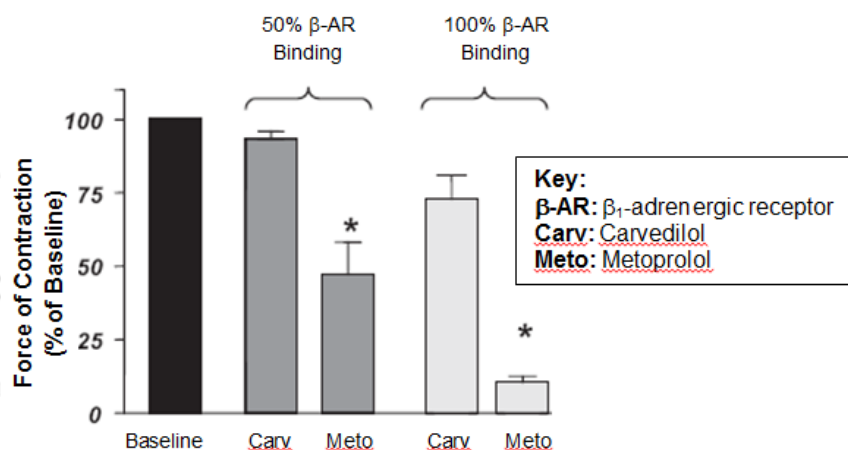


Fig. 4.2

(c) With reference to **Fig. 4.2**,

- (i) compare the effects of carvedilol and metoprolol on heart muscle contraction. [2]
- (ii) predict the effect of administering metoprolol to cardiac patients with tachycardia (increased and unstable heart rate) and hypertension (high blood pressure). [2]

The gene sequence of a newly discovered gene hypothesized to encode a fourth subtype of the β -adrenergic receptor has been determined. The protein sequences of the three β -adrenergic receptor subtypes have also been determined experimentally.

- (d)(i) Describe how you would use a bioinformatics tool to determine whether this new gene encodes a fourth subtype of the β -adrenergic receptor. [2]
- (ii) Suggest why the search algorithm described in (d)(i) may not yield a protein that exists in reality. [1]

[Total: 10 marks]

QUESTION 5

Fatty acid synthase (FAS) is a multi-enzyme complex that plays a role in fatty acid synthesis from acetyl coA. The synthesis of fatty acids requires repeated addition of two carbon acetyl coA units until the fatty acid chain reaches 16 carbons in length.

Two known classes of FAS exist, FAS I and FAS II. FAS II consists of multiple individual enzymes that carry out the different chemical reactions required for the synthesis of fatty acids, whereas FAS I is composed of two identical subunits that are each multi-functional. In each subunit of FAS I, three catalytic domains are located at the N-terminal end, separated from four other catalytic domains at the C-terminal end by a central core region of approximately 600 amino acid residues. The various domains of FAS I are shown in **Fig. 5**.

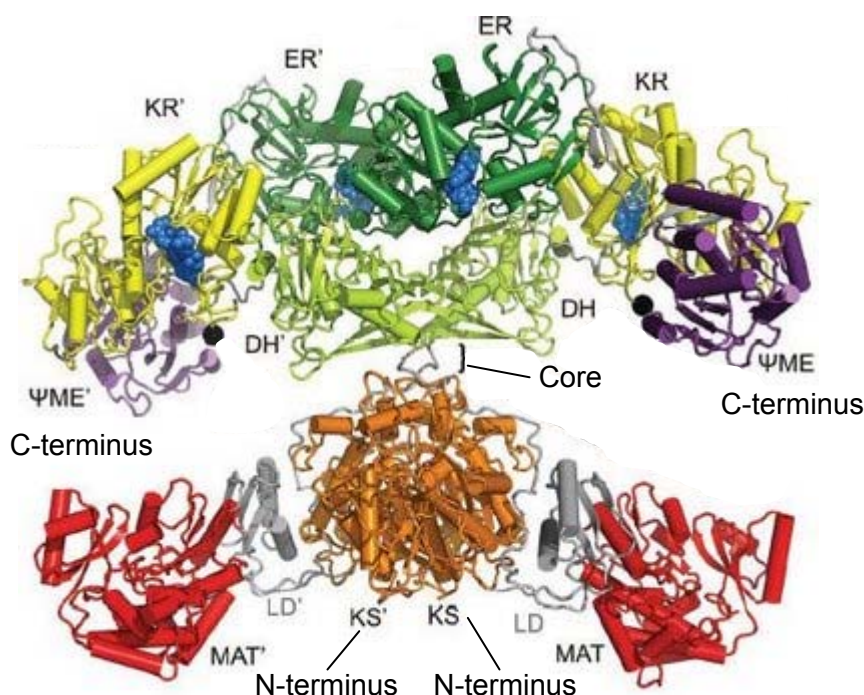


Fig. 5

- (a) Distinguish between the structures of FAS I and the pyruvate dehydrogenase complex. [2]
- (b) With reference to **Fig. 5**, state the highest level of protein structure attained by FAS I. [1]
- (c) State and explain whether the rate of fatty acid synthesis would be higher in FAS I or FAS II. [2]
- (d) Suggest two reasons why a central core region is necessary to separate the three N-terminal catalytic domains from the four C-terminal catalytic domains. [2]

FAS activity can be downregulated by cAMP-dependent protein kinase that carries out phosphorylation of the acetyl coA carboxylase catalytic domain, which is responsible for the conversion of acetyl coA to malonyl coA. This conversion is the first step in the catalytic pathway. cAMP-dependent protein kinase is activated in the presence of glucagon.

- (e) Discuss why it would be advantageous to the cell to carry out such regulation. [3]

[Total: 10 marks]

-- End of Section A --

SECTION B: STRUCTURED QUESTIONS

Answer any **THREE** questions in this section.

QUESTION 6

- (a) Fig. 6.1 shows the effect of pH on the net charge of a protein which has its isoelectric point (pI) at pH 8.

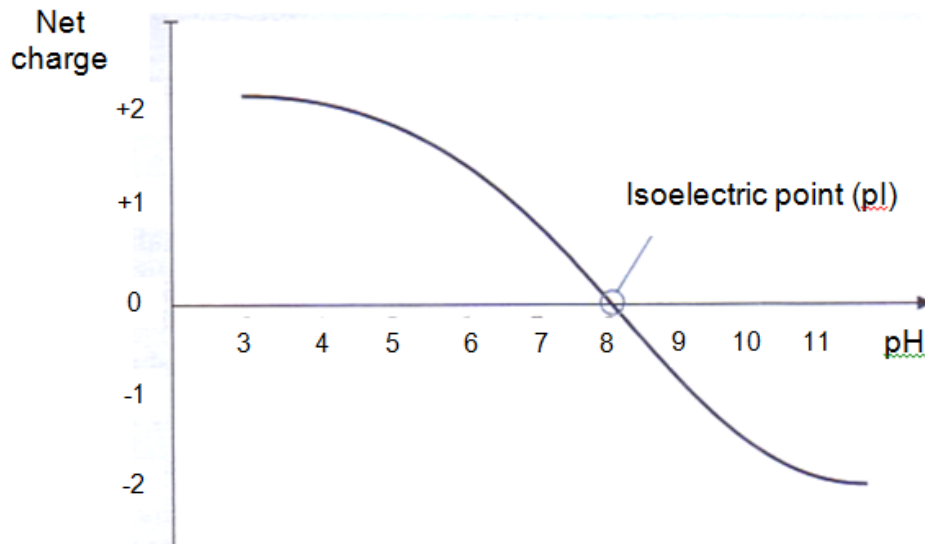


Fig. 6.1

State what is meant by the isoelectric point of a protein and with reference to Fig. 6.1, explain the change in the charge of a protein as pH increases. [3]

- (b) Targeting or signal sequences are important in both cotranslational translocation and posttranslational translocation. Give one difference between these two mechanisms of protein translocation, and suggest the fate of a protein that does not have a targeting signal. [2]
- (c) Titin, a muscle protein with a relative molecular mass of 3×10^6 , is the largest polypeptide yet described. Titin is composed of a large number of immunoglobulin (Ig) domains. The structure of an Ig domain is shown in Fig. 6.2.



Fig. 6.2

- (i) The Ig domain is characteristically found in antibody molecules. Account for the occurrence of the Ig domain in repeated copies across such functionally diverse proteins as titin and antibody molecules. [3]

Titin molecules extend from muscle thick filaments to the Z disc in the sarcomere, as illustrated in Fig. 6.3.

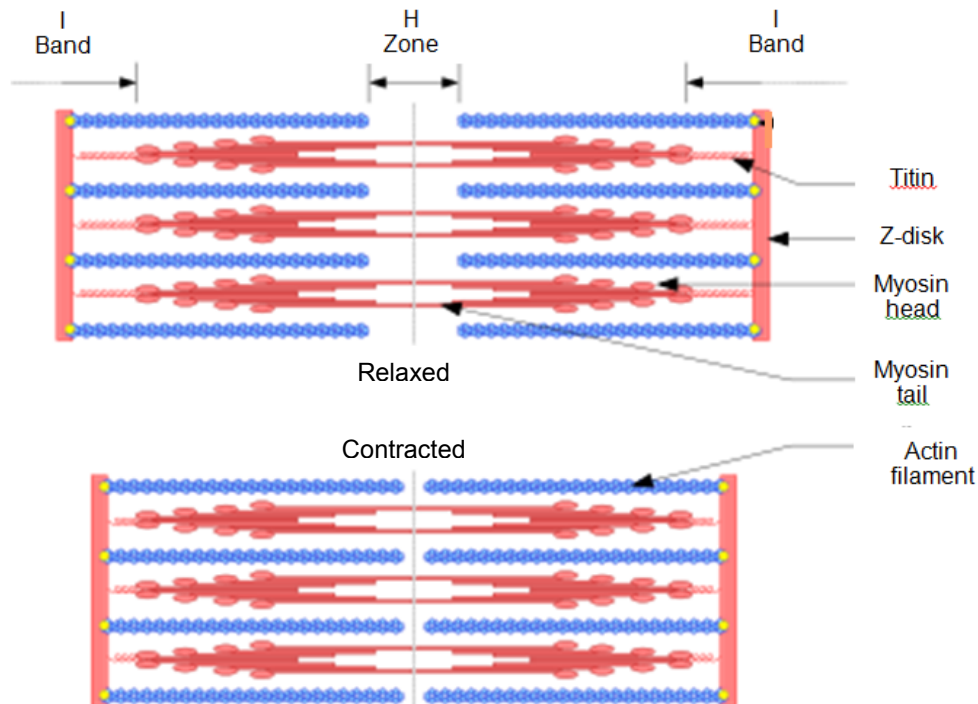


Fig. 6.3

- (ii) The Ig domains in the titin protein unfold when the protein is stretched during muscle relaxation and refold when the tension is removed during muscle contraction. Suggest a possible role of titin in muscle, and how the structure of the Ig domain contributes to this function. [2]

[Total: 10 marks]

QUESTION 7

Liver cells secrete a broad spectrum of proteins into the blood. Experiments are conducted to identify the time taken for different proteins to be secreted. A radioactive amino acid is added to cultured liver cells to label proteins as they are being synthesised.

- (a) Suggest, with a brief explanation, a suitable amino acid to be used for the experiment. [1]

The medium is then sampled at various times to measure the appearance of individual labelled proteins. The results are shown in **Fig. 7.1**.

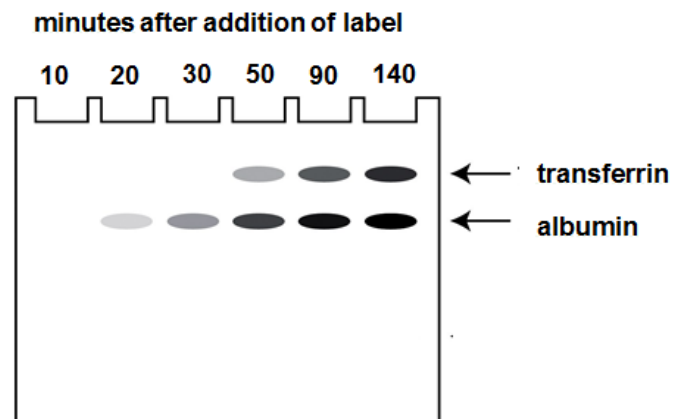


Fig. 7.1

(b) With reference to **Fig. 7.1**,

(i) describe how these proteins are separated and visualised. [3]

(ii) state the time taken for the first appearance of transferrin and albumin. [1]

Scientists were puzzled over the variability in secretion rates of different proteins. A further experiment is carried to determine the slow steps in the secretory pathway, which was hypothesised to occur in the endoplasmic reticulum (ER) or the Golgi apparatus (GA).

(c) Suggest why the scientists believe that the slow steps occur in either the ER or GA. [1]

To determine which organelle is the site of the occurrence of the slow steps, cells were labelled for four hours. The cells were then homogenised to break the ER and GA into vesicles before centrifugation was carried out. The amount of labelled albumin and transferrin that are associated with the two types of vesicles were then measured. The results are shown in **Fig. 7.2**.

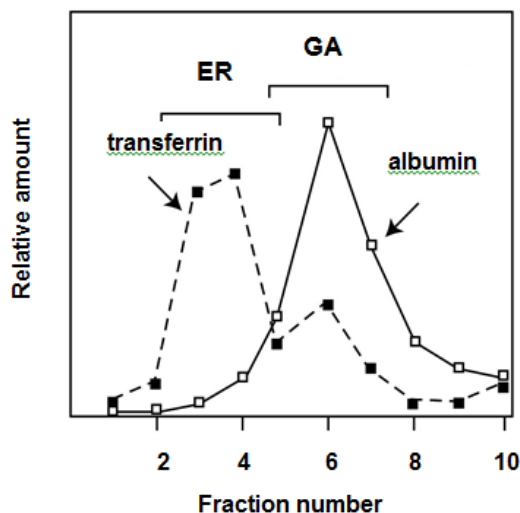


Fig. 7.2

(d) With reference to Fig. 7.2,

(i) briefly describe the type of centrifugation used. [2]

(ii) describe in which organelles the slow steps occur in the secretion of transferrin and albumin respectively. [2]

[Total: 10 marks]

QUESTION 8

Liver fatty acid-binding proteins (L-FABP) are a member of a group of proteins that play an active role in fatty acid-mediated signal transduction pathways and have thus been implicated in tumourigenesis. L-FABP expression profiles were generated from normal individuals and colorectal cancer patients, as shown in Fig. 8.

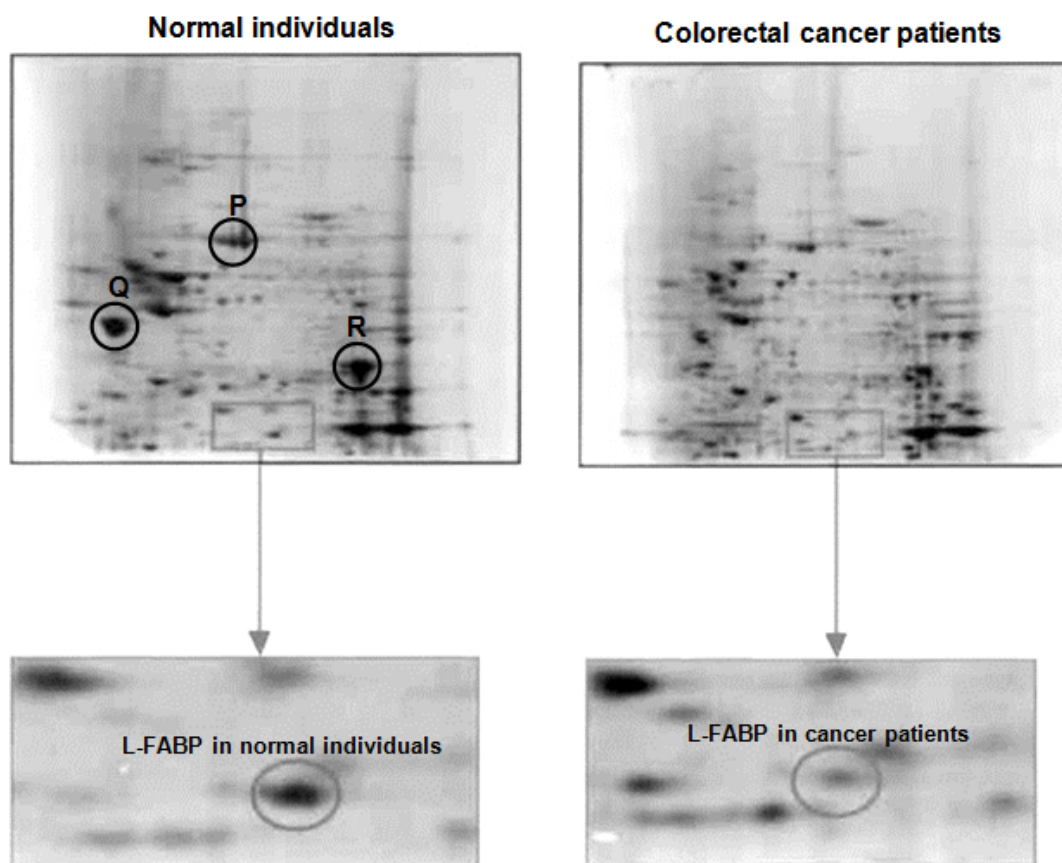


Fig. 8

- (a) Describe how the protein expression profile is obtained. [3]
- (b) With reference to **Fig. 8**, given that the molecular weights of proteins **P** and **Q** are 87kDa and 25kDa respectively, calculate the molecular weight of protein **R**. [2]
- (c) With reference to **Fig. 8**,
- (i) deduce if L-FABP is a tumour suppressor gene or proto-oncogene, and explain your answer. [2]
 - (ii) explain how L-FABP is implicated in colorectal cancer. [2]
- (d) State what subsequent method must be employed to determine the identities of proteins **P**, **Q** and **R**. [1]

[Total: 10 marks]

QUESTION 9

Specific nucleotide sequences on DNA double helix can serve as recognition sites for the binding of specific gene regulatory proteins such as transcription factors. These transcription factors contain structural features that allow them to recognise these specific nucleotide sequences.

- (a) (i) Name one possible motif these regulatory proteins could have in common. [1]
- (ii) Describe the structural features of the motif named in (a)(i). [2]
- (b) The amino acid sequences of transcription factor **1**, **2** and **3** which occur in organisms **X**, **Y** and **Z** respective are shown in **Table 9**.
- (i) With reference to the **Table 9**, calculate the score for alignments of the amino acid sequences using only the first 30 amino acids (assume a score of +2 for identical amino acids, a score of +1 for amino acids with similar properties and a score of -1 for amino acids with different properties). [3]

Transcription Factor 1		
MASGWANDSA	CKKAFDATID	LIQARARNTG
Transcription Factor 2		
MASGWANDSA	VQEQIDATID	DAIARARNGM
Transcription Factor 3		
MASLIKFLCW	VQEQIDATID	AQNARARNGM

Table 9

- (ii) Explain which two organisms are evolutionarily more closely related. [2]
- (iii) Explain why it is possible that your conclusion in (b)(ii) could be incorrect. [1]
- (iv) Name one programme that can be used to help in the construction of a phylogenetic tree. [1]

[Total: 10 marks]**-- End of Section B --**

SECTION C: FREE RESPONSE QUESTION

Answer the question in this section.

QUESTION 10

- (a) Describe how lysozyme interacts with its substrate. [6]
- (b) Distinguish between the structure and function of protein kinase A and tyrosine kinase receptors. [8]
- (c) Describe the use of Edman degradation in determining a new member of the protein kinase A family, and discuss the need for both a complementary and alternative technique. [6]

[Total: 20 marks]

-- End of Section C --

-- End of Paper --

TABLE OF AMINO ACIDS

Nonpolar						
	Glycine (Gly)	Alanine (Ala)	Valine (Val)	Leucine (Leu)	Isoleucine (Ile)	
Methionine (Met)	Phenylalanine (Phe)	Tryptophan (Trp)	Proline (Pro)			
Polar						
	Serine (Ser)	Threonine (Thr)	Cysteine (Cys)	Tyrosine (Tyr)	Asparagine (Asn)	Glutamine (Gln)
Electrically charged	Acidic			Basic		
	Aspartic acid (Asp)	Glutamic acid (Glu)	Lysine (Lys)	Arginine (Arg)	Histidine (His)	